

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If $p(x) = g(x) \cdot q(x) + r(x)$, and $p(x) = x^2 + 3x + 2$, $g(x) = x + 1$, then the quotient $q(x)$ is:
(A) (A) $x + 2$
(B) (B) $x - 2$
(C) (C) $x + 1$
(D) (D) $x - 1$
2. According to the division algorithm for polynomials, if $p(x)$ is divided by $g(x)$, then the remainder $r(x)$ must satisfy:
(A) (A) $\deg(r(x)) < \deg(g(x))$
(B) (B) $\deg(r(x)) = \deg(g(x))$
(C) (C) $\deg(r(x)) \geq \deg(g(x))$
(D) (D) $r(x) = 1$
3. If $p(x) = x^2 - 4$ is divided by $g(x) = x - 2$, what is the remainder?
(A) (A) 4
(B) (B) 2
(C) (C) -4
(D) (D) 0
4. Given $p(x) = 2x^2 + 5x + 3$ and $g(x) = 2x + 3$, the quotient $q(x)$ is:
(A) (A) $x - 1$
(B) (B) $x + 1$
(C) (C) $2x + 1$
(D) (D) $x + 3$
5. If the dividend is $x^2 + 5x + 6$, the divisor is $x + 2$, and the remainder is 0, what is the quotient?

- (A) (A) $x + 2$
 (B) (B) $x - 3$
 (C) (C) $x + 3$
 (D) (D) $x - 2$
6. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:
 (A) (A) 10 cm
 (B) (B) 11 cm
 (C) (C) 12 cm
 (D) (D) 15 cm
7. If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then $\angle POA$ is equal to:
 (A) (A) 50°
 (B) (B) 60°
 (C) (C) 70°
 (D) (D) 80°
8. The number of tangents that can be drawn from a point lying inside a circle is:
 (A) (A) 0
 (B) (B) 1
 (C) (C) 2
 (D) (D) Infinite
9. Assertion (A): For a polynomial $p(x) = x^2 - 3x + 2$ and divisor $g(x) = x - 1$, the remainder is 0. Reason (R): If $g(x)$ is a factor of $p(x)$, then $p(x) = g(x)q(x) + r(x)$, where $r(x) = 0$.
 (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true
10. Assertion (A): The degree of the remainder $r(x)$ must be less than the degree of the divisor $g(x)$. Reason (R): In the division algorithm $p(x) = g(x)q(x) + r(x)$, $r(x)$ can be a non-zero constant even if $g(x)$ has degree 1.
 (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
11. Assertion (A): If $p(x) = x^2 + 1$ is divided by $g(x) = x$, the remainder is 1. Reason (R): The remainder of $p(x)$ divided by $x - a$ is $p(a)$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
12. Assertion (A): If $p(x) = 2x^2 + 4x + 2$ is divided by $g(x) = x + 1$, the quotient is $2x + 2$. Reason (R): $2x^2 + 4x + 2 = (x + 1)(2x + 2) + 0$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
13. Assertion (A): The division algorithm for polynomials is applicable only when the divisor $g(x) \neq 0$. Reason (R): Division by zero is undefined in mathematics.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

- On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $x - 2$ and $-2x + 4$ respectively. Find $g(x)$.
- Check whether $x^2 + 3x + 1$ is a factor of $3x^4 + 5x^3 - 7x^2 + 2x + 2$ by applying the division algorithm.
- Find the quotient and remainder when $p(x) = x^3 - 3x^2 + 5x - 3$ is divided by $g(x) = x^2 - 2$.

- What must be subtracted from $f(x) = x^4 + 2x^3 - 2x^2 + 4x - 1$ so that the resulting polynomial is exactly divisible by $x^2 + 2$?
- If the polynomial $x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $x^2 - 2x + k$, the remainder comes out to be $x + a$. Find the values of k and a .

Section C: Short Answer (3 marks each)

- The mean of the following frequency distribution is 50. Find the value of p if the sum of frequencies is 100: — Class — 0-20 — 20-40 — 40-60 — 60-80 — 80-100 —
— Frequency — 17 — p — 32 — p — 19 —
- The following distribution shows the daily pocket allowance of children of a locality. If the mean pocket allowance is Rs. 18, find the missing frequency f : — Daily allowance (in Rs) — 11-13 — 13-15 — 15-17 — 17-19 — 19-21 — 21-23 — 23-25 — —
— Number of children — 7 — 6 — 9 — 13 — f — 5
— 4 —
- Find the median of the following data: — Marks — 0-10 — 10-20 — 20-30 — 30-40 — 40-50 — —
— Number of students — 5 — 15 — 20 — 7 — 3 —
- The following table shows the ages of 100 patients admitted in a hospital during a year. Find the mode of the ages: — Age (in years) — 5-15 — 15-25 — 25-35 — 35-45 — 45-55 — 55-65 — —
— Number of patients — 6 — 11 — 21 — 23 — 14 — 5 —
- A survey conducted on 20 households in a locality by a group of students resulted in the following frequency table for the number of family members in a household: — Family size — 1-3 — 3-5 — 5-7 — 7-9 — 9-11 — —
Frequency — 7 — 8 — 2 — 2 — 1 — Find the mode of this data.
- On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $x - 2$ and $-2x + 4$, respectively. Find $g(x)$.
- Find the values of a and b if the polynomial $x^4 + x^3 + 8x^2 + ax + b$ is exactly divisible by $x^2 + 1$.
- Divide the polynomial $p(x) = 3x^2 - x^3 - 3x + 5$ by $g(x) = x - 1 - x^2$ and verify the division algorithm.
- What must be subtracted from $8x^4 + 14x^3 - 2x^2 + 7x - 8$ so that the resulting polynomial is exactly divisible by $4x^2 + x - 2$?
- Obtain all zeros of $3x^4 + 6x^3 - 2x^2 - 10x - 5$, if two of its zeros are $\sqrt{5/3}$ and $-\sqrt{5/3}$.

Section D: Long Answer (4-5 marks each)

- The mean of the following frequency distribution is 50. Find the values of p and q , if the sum of all frequencies is 120.

Class Interval	0-20	20-40	40-60	60-80	80-100	—	:	—	:	—	:	—
Frequency	17	p	32	q	19	—						
- The median of the following data is 525. Find the values of x and y , if the total frequency is 100.

Class Interval	0-100	100-200	200-300	300-400	400-500	500-600	600-700	700-800	800-900	900-1000	—	:	—	:	—	:	—	:	—	:	—
Frequency	2	5	x	12	17	20	y	9	7	4	—										
- The following distribution gives the daily income of 50 workers of a factory. Convert the distribution above to a 'less than type' cumulative frequency distribution and draw its ogive. From the ogive, find the median daily income.

Daily Income (in Rs.)	100-120	120-140	140-160	160-180	180-200	—	:	—	:	—	:	—	:	—	:	—	:	—	:	—
Number of workers	12	14	8	6	10	—														
- Find the mode of the following frequency distribution. If the mean of the same data is 28.6, verify the empirical relationship between mean, median, and mode.

Class	0-10	10-20	20-30	30-40	40-50	—	:	—	:	—	:	—	:	—	:	—	:	—	:	—
Frequency	5	8	20	15	7	—														
- Obtain all zeroes of the polynomial $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$, if two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.
- Find the values of a and b so that the polynomial $x^4 + x^3 + 8x^2 + ax + b$ is exactly divisible by $x^2 + 1$.
- If the polynomial $f(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $g(x) = x^2 - 2x + k$, the remainder comes out to be $x + a$. Find the values of k and a .
- A company models its profit $P(x)$ in thousands of rupees by the polynomial $P(x) = -x^3 + 4x^2 + x - 4$, where x represents the number of units sold. If the factor representing the price per unit is $(x - 4)$, find the other factors of the profit polynomial.
- Prove that the lengths of tangents drawn from an external point to a circle are equal.
- A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.
- Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.
- If a hexagon ABCDEF circumscribes a circle, prove that $AB + CD + EF = BC + DE + FA$.